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## Microelectronic devices, related electronic systems, and methods of forming microelectronic devices

### Abstract


A microelectronic device comprises vertical stacks of memory cells, each vertical stack of memory cells comprising a vertical stack of access devices, a vertical stack of capacitors horizontally neighboring the vertical stack of access devices, and a conductive pillar structure vertically extending through the vertical stack of access devices. The microelectronic device further comprises multiplexers and additional transistors vertically overlying the vertical stacks of memory cells, and global digit lines vertically overlying the multiplexer and the additional transistor. Related electronic systems and methods are also described.



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## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 5901099 | 12/1998 | Gilliam et al. | N/A | N/A |
| 6778452 | 12/2003 | Beigel et al. | N/A | N/A |
| 8067286 | 12/2010 | Parekh et al. | N/A | N/A |
| 8153499 | 12/2011 | Or-Bach et al. | N/A | N/A |
| 8184471 | 12/2011 | Woo et al. | N/A | N/A |
| 8546188 | 12/2012 | Liu et al. | N/A | N/A |
| 8716116 | 12/2013 | Parekh et al. | N/A | N/A |
| 9543433 | 12/2016 | Anathan et al. | N/A | N/A |
| 10163480 | 12/2017 | Kawamura et al. | N/A | N/A |
| 10366740 | 12/2018 | Kim et al. | N/A | N/A |
| 11848309 | 12/2022 | Simsek-Ege | N/A | N/A |
| 2011/0169072 | 12/2010 | Lim | 257/E21.409 | H10B 43/27 |
| 2014/0347945 | 12/2013 | Venkata et al. | N/A | N/A |
| 2018/0323199 | 12/2017 | Roberts | N/A | H10D 64/518 |
| 2019/0074277 | 12/2018 | Ramaswamy | N/A | N/A |
| 2019/0244656 | 12/2018 | Li | N/A | G11C 11/4097 |
| 2019/0244933 | 12/2018 | Or-Bach et al. | N/A | N/A |
| 2020/0052070 | 12/2019 | Derner | N/A | H10B 12/50 |
| 2020/0066730 | 12/2019 | Guo et al. | N/A | N/A |
| 2020/0105330 | 12/2019 | Kimura | N/A | H10D 87/00 |
| 2021/0217460 | 12/2020 | Montierth et al. | N/A | N/A |
| 2022/0399308 | 12/2021 | Simsek-Ege et al. | N/A | N/A |
| 2023/0018127 | 12/2022 | Gandhi et al. | N/A | N/A |
| 2023/0207505 | 12/2022 | Simsek-Ege et al. | N/A | N/A |
| 2023/0262988 | 12/2022 | Harari | 257/295 | H10D 64/033 |

**OTHER PUBLICATIONS**

Simsek-Ege et al., “Microelectronic Devices, Related Electronic Systems, and Methods of Forming Microelectronic Devices”, filed May 26, 2022, U.S. Appl. No. 17/804,234, 77 pages. cited by applicant

Simsek-Ege, “Microelectronic Devices, Related Electronic Systems, and Methods of Forming Microelectronic Devices”, filed May 26, 2022, U.S. Appl. No. 17/804,247, 89 pages. cited by applicant

Simsek-Ege, “Microelectronic Devices, Related Electronic Systems, and Methods of Forming Microelectronic Devices”, filed May 26, 2022, U.S. Appl. No. 17/804,258, 98 pages. cited by applicant

Simsek-Ege, “Microelectronic Devices, Related Electronic Systems, and Methods of Forming

Microelectronic Devices”, filed May 26, 2022, U.S. Appl. No. 17/804,270, 119 pages. cited by applicant

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## Background/Summary

TECHNICAL FIELD

(1) The disclosure, in various embodiments, relates generally to the field of microelectronic device design and fabrication. More specifically, the disclosure relates to methods of forming microelectronic devices from independently formed microelectronic device structures, and to related microelectronic devices and electronic systems.

BACKGROUND

(2) Microelectronic device designers often desire to increase the level of integration or density of features within a microelectronic device by reducing the dimensions of the individual features and by reducing the separation distance between neighboring features. In addition, microelectronic device designers often desire to design architectures that are not only compact, but offer performance advantages, as well as simplified designs.

(3) One example of a microelectronic device is a memory device. Memory devices are generally provided as internal integrated circuits in computers or other electronic devices. There are many types of memory devices including, but not limited to, volatile memory devices, such as dynamic random access memory (DRAM) devices; and non-volatile memory devices such as NAND Flash memory devices. A typical memory cell of a DRAM device includes one access device, such as a transistor, and one memory storage structure, such as a capacitor. Modern applications for semiconductor devices can employ significant quantities of memory cells, arranged in memory arrays exhibiting rows and columns of the memory cells. The memory cells may be electrically accessed through digit lines (e.g., bit lines, data lines) and word lines (e.g., access lines) arranged along the rows and columns of the memory cells of the memory arrays. Memory arrays can be two-dimensional (2D) so as to exhibit a single deck (e.g., a single tier, a single level) of the memory cells, or can be three-dimensional (3D) so as to exhibit multiple decks (e.g., multiple levels, multiple tiers) of the memory cells.

(4) Control logic devices within a base control logic structure underlying a memory array of a memory device have been used to control operations (e.g., access operations, read operations, write operations) of the memory cells of the memory device. An assembly of the control logic devices may be provided in electrical communication with the memory cells of the memory array by way of routing and interconnect structures. However, processing conditions (e.g., temperatures, pressures, materials) for the formation of the memory array over the base control logic structure can limit the configurations and performance of the control logic devices within the base control logic structure. In addition, the quantities, dimensions, and arrangements of the (Efferent control devices employed within the base control logic structure can also undesirably impede reductions to the size e a horizontal footprint) of the memory device, and/or improvements in the performance (e.g., faster memory cell ON/OFF speed, lower threshold switching voltage requirements, faster data transfer rates, lower power consumption) of the memory device. Furthermore, as the density and complexity of the memory array have increased, so has the complexity of the control logic

devices. In some instances, the control logic devices consume more real estate than the memory devices, reducing the memory density of the memory device.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A through FIG. **1**L are simplified partial top-down views (FIG. **1**A and FIG. **1**J) and simplified partial cross-sectional views (FIG. **1**B through FIG. **1**I, FIG. **1**K, and FIG. **1**L) illustrating a method of forming a first microelectronic device structure, in accordance with embodiments of the disclosure;

(2) FIG. **2**A and FIG. **2**B are simplified partial cross-sectional views of a microelectronic device formed from the first microelectronic device structure and a second microelectronic device structure, in accordance with embodiments of the disclosure; and

(3) FIG. **3** is a schematic block diagram of an electronic system, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

(4) The illustrations included herewith are not meant to be actual views of any particular systems, microelectronic structures, microelectronic devices, or integrated circuits thereof, but are merely idealized representations that are employed to describe embodiments herein. Elements and features common between figures may retain the same numerical designation except that, for ease of following the description, reference numerals begin with the number of the drawing on which the elements are introduced or most fully described.

(5) The following description provides specific details, such as material types, material thicknesses, and processing conditions in order to provide a thorough description of embodiments described herein. However, a person of ordinary skill in the art will understand that the embodiments disclosed herein may be practiced without employing these specific details. Indeed, the embodiments may be practiced in conjunction with conventional fabrication techniques employed in the semiconductor industry. In addition, the description provided herein does not form a complete process flow for manufacturing a microelectronic device (e.g., a semiconductor device, a memory device), apparatus, or electronic system, or a complete microelectronic device, apparatus, or electronic system. The structures described below do not form a complete microelectronic device, apparatus, or electronic system. Only those process acts and structures necessary to understand the embodiments described herein are described in detail below. Additional acts to form a complete microelectronic device, apparatus, or electronic system from the structures may be performed by conventional techniques.

(6) The materials described herein may be formed by conventional techniques including, but not limited to, spin coating, blanket coating, chemical vapor deposition (CVD), atomic layer deposition (ALD), plasma enhanced ALD, physical vapor deposition (PVD), plasma enhanced chemical vapor deposition (PECVD), or low pressure chemical vapor deposition (LPCVD). Alternatively, the materials may be grown in situ. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. The removal of materials may be accomplished by any suitable technique including, but not limited to, etching, abrasive planarization (e.g., chemical-mechanical planarization), or other known methods unless the context indicates otherwise.

(7) As used herein, the term “configured” refers to a size, shape, material composition, orientation, and arrangement of one or more of at least one structure and at least one apparatus facilitating operation of one or more of the structure and the apparatus in a predetermined way.

(8) As used herein, the terms “longitudinal,” “vertical,” “lateral,” and “horizontal” are in reference to a major plane of a substrate (e.g., base material, base structure, base construction, etc.) in or on

which one or more structures and/or features are formed and are not necessarily defined by Earth's gravitational field. A “lateral” or “horizontal” direction is a direction that is substantially parallel to the major plane of the substrate, while a “longitudinal” or “vertical” direction is a direction that is substantially perpendicular to the major plane of the substrate. The major plane of the substrate is defined by a surface of the substrate having a relatively large area compared to other surfaces of the substrate. With reference to the figures, a “horizontal” or “lateral” direction may be perpendicular to an indicated “Z” axis, and may be parallel to an indicated “X” axis and/or parallel to an indicated “Y” axis; and a “vertical” or “longitudinal” direction may be parallel to an indicated “Z” axis, may be perpendicular to an indicated “X” axis, and may be perpendicular to an indicated “Y” axis.
(9) As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a degree of variance, such as within acceptable tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90.0 percent met, at least 95.0 percent met, at least 99.0 percent met, at least 99.9 percent met, or even 100.0 percent met.
(10) As used herein, “about” or “approximately” in reference to a numerical value for a particular parameter is inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately” in reference to a numerical value may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.
(11) As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “bottom,” “above,” “upper,” “top,” “front,” “rear,” “left,” “right,” and the like, may be used for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of the materials in addition to the orientation depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “beneath” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (e.g., rotated 90 degrees, inverted, flipped, etc.) and the spatially relative descriptors used herein interpreted accordingly.
(12) As used herein, features (e.g., regions, materials, structures, devices) described as “neighboring” one another means and includes features of the disclosed identity (or identities) that are located most proximate (e.g., closest to) one another. Additional features (e.g., additional regions, additional materials, additional structures, additional devices) not matching the disclosed identity (or identities) of the “neighboring” features may be disposed between the “neighboring” features. Put another way, the “neighboring” features may be positioned directly adjacent one another, such that no other feature intervenes between the “neighboring” features; or the “neighboring” features may be positioned indirectly adjacent one another, such that at least one feature having an identity other than that associated with at least one the “neighboring” features is positioned between the “neighboring” features. Accordingly, features described as “vertically neighboring” one another means and includes features of the disclosed identity (or identities) that are located most vertically proximate (e.g., vertically closest to) one another. Moreover, features described as “horizontally neighboring” one another means and includes features of the disclosed

identity (or identities) that are located most horizontally proximate (e.g., horizontally closest to) one another.

(13) As used herein, the term “memory device” means and includes microelectronic devices exhibiting memory functionality, but not necessarily limited to memory functionality. Stated another way, and by way of example only, the term “memory device” means and includes not only conventional memory (e.g., conventional volatile memory, such as conventional DRAM; conventional non-volatile memory, such as conventional NAND memory), but also includes an application specific integrated circuit (ASIC) (e.g., a system on a chip (SoC)), a microelectronic device combining logic and memory, and a graphics processing unit (GPU) incorporating memory.

(14) As used herein, “conductive material” means and includes electrically conductive material such as one or more of a metal (e.g., tungsten (W), titanium (Ti), molybdenum (Mo), niobium (Nb), vanadium (V), hafnium (Hf), tantalum (Ta), chromium (Cr), zirconium (Zr), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Jr), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), aluminum (Al)), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a magnesium (Mg)-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide), and a conductively doped semiconductor material (e.g., conductively doped polysilicon, conductively doped germanium (Ge), conductively doped silicon germanium (SiGe)). In addition, a “conductive structure” means and includes a structure formed of and including a conductive material.

(15) As used herein, “insulative material” means and includes electrically insulative material, such one or more of at least one dielectric oxide material (e.g., one or more of a silicon oxide (SiO.sub.x), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, an aluminum oxide (AlO.sub.x), a hafnium oxide (HfO.sub.x), a niobium oxide (NbO.sub.x), a titanium oxide (TiO.sub.x), a zirconium oxide (ZrO.sub.x), a tantalum oxide (TaO.sub.x), and a magnesium oxide (MgO.sub.x)), at least one dielectric nitride material (e.g., a silicon nitride (SiN.sub.y)), at least one dielectric oxynitride material (e.g., a silicon oxynitride (SiO.sub.xN.sub.y)), and at least one dielectric carboxynitride material (e.g., a silicon carboxynitride (SiO.sub.xC.sub.zN.sub.y)). Formulae including one or more of “x,” “y,” and “z” herein (e.g., SiO.sub.x, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, SiN.sub.y, SiO.sub.xN.sub.y, SiO.sub.xC.sub.zN.sub.y) represent a material that contains an average ratio of “x” atoms of one element, “y” atoms of another element, and “z” atoms of an additional element (if any) for every one atom of another element (e.g., Si, Al, Hf, Nb, Ti). As the formulae are representative of relative atomic ratios and not strict chemical structure, an insulative material may comprise one or more stoichiometric compounds and/or one or more non-stoichiometric compounds, and values of “x,” “y,” and “z” (if any) may be integers or may be non-integers. As used herein, the term “non-stoichiometric compound” means and includes a chemical compound with an elemental composition that cannot be represented by a ratio of well-defined natural numbers and is in violation of the law of definite proportions. In addition, an “insulative structure” means and includes a structure formed of and including an insulative material.

(16) As used herein, “semiconductor material” or “semiconductive material” refers to a material having an electrical conductivity between those of insulative materials and conductive materials. For example, a semiconductor material may have an electrical conductivity of between about 10.sup.−8 Siemens per centimeter (S/cm) and about 10.sup.4 S/cm (10.sup.6 S/m) at room temperature. Examples of semiconductor materials include elements found in column IV of the periodic table of elements such as silicon (Si), germanium (Ge), and carbon (C). Other examples of semiconductor materials include compound semiconductor materials such as binary compound semiconductor materials (e.g., gallium arsenide (GaAs)), ternary compound semiconductor

materials (e.g., Al.sub.XGa.sub.1-XAs), and quaternary compound semiconductor materials (e.g., Ga.sub.XIn.sub.1-XAs.sub.YP.sub.1-Y), without limitation. Compound semiconductor materials may include combinations of elements from columns III and V of the periodic table of elements (III-V semiconductor materials) or from columns II and VI of the periodic table of elements (II-VI semiconductor materials), without limitation. Further examples of semiconductor materials include oxide semiconductor materials such as zinc tin oxide (Zn.sub.xSn.sub.yO, commonly referred to as "ZTO"), indium zinc oxide (In.sub.xZn.sub.yO, commonly referred to as "IZO"), zinc oxide (Zn.sub.xO), indium gallium zinc oxide (In.sub.xGa.sub.yZn.sub.zO, commonly referred to as "IGZO"), indium gallium silicon oxide (In.sub.xGa.sub.ySi.sub.zO, commonly referred to as "IGSO"), indium tungsten oxide (In.sub.xW.sub.yO, commonly referred to as "IWO"), indium oxide (In.sub.xO), tin oxide (Sn.sub.xO), titanium oxide (Ti.sub.xO), zinc oxide nitride (Zn.sub.xON.sub.z), magnesium zinc oxide (Mg.sub.xZn.sub.yO), zirconium indium zinc oxide (Zr.sub.xIn.sub.yZn.sub.zO), hafnium indium zinc oxide (Hf.sub.xIn.sub.yZn.sub.zO), tin indium zinc oxide (Sn.sub.xIn.sub.yZn.sub.zO), aluminum tin indium zinc oxide (Al.sub.xSn.sub.yIn.sub.zZn.sub.aO), silicon indium zinc oxide (Si.sub.xIn.sub.yZn.sub.zO), aluminum zinc tin oxide (Al.sub.xZn.sub.ySn.sub.zO), gallium zinc tin oxide (Ga.sub.xZn.sub.ySn.sub.zO), zirconium zinc tin oxide (Zr.sub.xZn.sub.ySn.sub.zO), and other similar materials.

(17) According to embodiments described herein, a first microelectronic device structure is formed by forming vertical stacks of memory cells. Each vertical stack of memory cells individually comprises a vertical stack of capacitor structures horizontally neighboring a vertical stack of access devices. A conductive pillar structure is in electrical communication with the vertical stack of access devices and may vertically extend through or proximate the vertical stack of access devices. A conductive plate structure vertically extends proximate to and in electrical communication with electrodes of each of the capacitor structures of the vertical stack of capacitor structures. In some embodiments, an additional vertical stack of memory cells horizontally neighbors the conductive plate structure and comprises additional vertical stacks of capacitor structures in electrical communication with the conductive plate structure and additional vertical stacks of access devices in electrical communication with the additional vertical stacks of capacitor structures. Multiplexers and additional transistors (e.g., "bleeder" transistors) vertically overlie the vertical stacks of memory cells. The multiplexers are configured to selectively electrically couple the conductive pillar structure of individual vertical stacks of memory cells to a global digit line vertically overlying the vertical stacks of memory cells. The additional transistors are configured to provide a bias to unselected memory cells by electrically connecting the conductive plate structure to the conductive pillar structure in electrical communication with unselected memory cells. In some embodiments, a source region of the multiplexers is shared with a source region of the additional transistors.

(18) Forming the multiplexers and the additional transistors vertically over the vertical stacks of memory cells facilitates forming a larger density of memory cells in the microelectronic device structure. Conductive routing structures between the multiplexers and components of the vertical stacks of memory cells (e.g., between the multiplexers and a global digit line, between the multiplexers and the conductive pillar structures of the vertical stacks of memory cells, between the additional transistors and the conductive pillar structures of the vertical stacks of memory cells, and between the additional transistors and the conductive plate structure) facilitates electrically isolating horizontally neighboring conductive pillar structures of horizontally neighboring vertical stacks of memory cells while also providing a sufficient quantity of multiplexers and additional transistors vertically over the vertical stacks of memory cells facilitate control operations of the vertical stacks of memory cells. For example, electrically isolating horizontally neighboring conductive pillar structures of horizontally neighboring vertical stacks of memory cells facilitates an improved signal to noise ratio of sense amplifier devices that are formed in electrical

communication with the global digit lines that are, in turn, in electrical communication with the multiplexers.
(19) FIG. **1**A through FIG. **1**L are simplified partial top-down views (FIG. **1**A and FIG. **1**J) and simplified partial cross-sectional views (FIG. **1**B through FIG. **1**I, FIG. **1**K, and FIG. **1**L) illustrating a method of forming a first microelectronic device structure **100** (e.g., a memory device, such as a 3D DRAM memory device), in accordance with embodiments of the disclosure. With the description provided below, it will be readily apparent to one of ordinary skill in the art that the methods and structures described herein with reference to FIG. **1**A through FIG. **1**L may be used in various devices and electronic systems. The first microelectronic device structure **100** may also be referred to herein as a first die or a first wafer.
(20) FIG. **1**A is a simplified partial top-down view of the first microelectronic device structure **100**; FIG. **1**B is a simplified partial cross-sectional view of the first microelectronic device structure **100** taken through section line B-B of FIG. **1**A; and FIG. **1**C is a simplified partial cross-sectional view of the first microelectronic device structure **100** taken through section line C-C of FIG. **1**A and section line C-C of FIG. **1**B.
(21) Referring to FIG. **1**A, the first microelectronic device structure **100** includes an array region **101** (also referred to herein as a “memory array region”) and one or more peripheral regions **103** located external to the array region **101**. In some embodiments, the peripheral regions **103** horizontally (e.g., in at least X-direction) surround the array region **101**. In some embodiments, the peripheral regions **103** substantially surround all horizontal sides of the array region **101** in a first horizontal direction (e.g., the X-direction). In other embodiments, the peripheral regions **103** substantially surround all horizontal boundaries (e.g., an entire horizontal area) of the array region **101**.
(22) With reference to FIG. **1**A and FIG. **1**B, within the array region **101**, the first microelectronic device structure **100** includes a vertical (e.g., in the Z-direction) stack structure **105** from which a vertical stack of memory cells (e.g., vertical stack of memory cells **120** (FIG. **1**J through FIG. **1**L)) will be formed over a first insulative material **114**. The stack structure **105** includes a vertically alternating (e.g., in the Z-direction) sequence of a first material **116** and a second material **118**.
(23) The first insulative material **114** may be formed of and include insulative material. In some embodiments, the first insulative material **114** is formed of and includes insulative material, such as one or more of an oxide material (e.g., silicon dioxide (SiO.sub.2), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, titanium dioxide (TiO.sub.2), hafnium oxide (HfO.sub.2), zirconium dioxide (ZrO.sub.2), hafnium dioxide (HfO.sub.2), tantalum oxide (TaO.sub.2), magnesium oxide (MgO), aluminum oxide (Al.sub.2O.sub.3), or a combination thereof), and amorphous carbon. In some embodiments, the first insulative material **114** comprises silicon dioxide. Each level of the first insulative material **114** may individually include a substantially homogeneous distribution of the at least one insulating material, or a substantially heterogeneous distribution of the at least one insulating material. As used herein, the term “homogeneous distribution” means amounts of a material do not vary throughout different portions (e.g., different horizontal portions, different vertical portions) of a structure. Conversely, as used herein, the term “heterogeneous distribution” means amounts of a material vary throughout different portions of a structure. Amounts of the material may vary stepwise (e.g., change abruptly), or may vary continuously (e.g., change progressively, such as linearly, parabolically) throughout different portions of the structure. In some embodiments, each level of the first insulative material **114** exhibits a substantially homogeneous distribution of insulative material. In additional embodiments, at least one of the levels of the first insulative material **114** exhibits a substantially heterogeneous distribution of at least one insulative material. The levels of the first insulative material **114** may, for example, be formed of and include a stack (e.g., laminate) of at least two different insulative materials. Each of the levels of the first insulative material **114** may each be substantially planar, and may each individually exhibit a desired thickness.

(24) The first material **116** may be formed of and include, for example, a semiconductive material (e.g., silicon) or an oxide material (e.g., silicon dioxide). In some embodiments, the first material **116** comprises silicon, such as epitaxially grown silicon. In some embodiments, the first material **116** comprises monocrystalline silicon.
(25) The second material **118** may have a different material composition than the first material **116** and may exhibit an etch selectivity with respect to the first material **116**. The second material **118** may be formed of and include one or more of silicon germanium, polysilicon, a nitride material (e.g., silicon nitride (Si.sub.3N.sub.4)), or an oxynitride material (e.g., silicon oxynitride). In some embodiments, such as where the first material **116** comprises silicon, the second material **118** comprises silicon germanium, such as epitaxially grown silicon germanium. In other embodiments, such as where the first material **116** comprises silicon, the second material **118** comprises polysilicon. In yet other embodiments, such as where the first material **116** comprises silicon dioxide, the second material **118** comprises silicon nitride or silicon oxynitride.
(26) With continued reference to FIG. **1**B, access devices **130** may be formed within the levels of the first material **116**. The access devices **130** may comprise doped portions of the first material **116** to form channel regions **134**. The channel regions **134** may be doped with one or more of at least one N-type dopant, such as one or more of arsenic ions, phosphorous ions, and antimony ions. In other embodiments, the channel regions **134** are doped with at least one P-type dopant, such as boron ions. In some embodiments, the channel regions **134** of the access devices **130** are horizontally between (e.g., in the X-direction, in the Y-direction) a source region and a drain region of the access devices **130**. The access devices **130** vertically overlying (e.g., in the Z-direction) one another may be form a vertical stack of access devices **130**.
(27) In some embodiments, conductive structures **132** vertically overlie (e.g., in the Z-direction) and vertically underlie (e.g., in the Z-direction) each of the access devices **130**, such as the channel regions **134** of each of the access devices **130**. In some embodiments, the channel regions **134** are vertically surrounded by the conductive structures **132**. The conductive structures **132** may individually be referred to herein as "first conductive lines," "access lines," or "word lines." In some embodiments, vertically neighboring (e.g., in the Z-direction) conductive structures **132** between vertically neighboring (e.g., in the Z-direction) access devices **130** are spaced from each other by the second insulative material **119**.
(28) The second insulative material **119** may be formed of and include an insulative material that is different than, and that has an etch selectivity with respect to, the first material **116**. In some embodiments, the second insulative material **119** is formed of and includes one or more of the materials described above with reference to the first insulative material **114**. In some embodiments, the second insulative material **119** is formed of and includes an oxide material (e.g., silicon dioxide).
(29) The conductive structures **132** may extend horizontally (e.g., in the X-direction; FIG. **1**A) through first microelectronic device structure **100** as lines (e.g., word lines) and may each be configured to be operably coupled to a vertically neighboring (e.g., in the Z-direction) access device **130** (e.g., the channel region **134** of a neighboring access device **130**). In other words, a conductive structure **132** may be configured to be operably coupled to a vertically neighboring access device **130**.
(30) The conductive structures **132** vertically overlying (e.g., in the Z-direction) and within horizontal boundaries (e.g., in the X-direction, in the Y-direction) of one another may form a vertical stack structure **135** of conductive structures **132**. As described in further detail below, each vertical stack structure **135** of conductive structures **132** may intersect more than one (e.g., two, three, four) vertical stacks of memory cells (e.g., vertical stacks of memory cells **120** (FIG. **1**J through FIG. **1**L)).
(31) The conductive structures **132** may individually be formed of and include conductive material, such as, for example, one or more of a metal (e.g., tungsten, titanium, nickel, platinum, rhodium,

ruthenium, aluminum, copper, molybdenum, iridium, silver, gold), a metal alloy, a metal-containing material (e.g., metal nitrides, metal silicides, metal carbides, metal oxides), a material including at least one of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), titanium aluminum nitride (TiAlN), iridium oxide ($IrO_x$), ruthenium oxide ($RuO_x$), alloys thereof, a conductively doped semiconductor material (e.g., conductively doped silicon, conductively doped germanium, conductively doped silicon germanium, etc.), polysilicon, or other materials exhibiting electrical conductivity. In some embodiments, the conductive structures **132** individually comprise tungsten. In other embodiments, the conductive structures individually comprise copper.

(32) With combined reference to FIG. **1**B and FIG. **1**C, each of the access devices **130** is surrounded by the dielectric material **140**, which may also be referred to herein as a “gate dielectric material.” In some embodiments, the portion of the conductive structure **132** directly vertically neighboring (e.g., in the Z-direction) and located within horizontal boundaries (e.g., in the X-direction, in the Y-direction) of the dielectric material **140** may be referred to as a “gate electrode.” In some embodiments, the conductive structures **132** are separated from the access devices **130** by the dielectric material **140**.

(33) In some embodiments, and as illustrated in FIG. **1**C, each of the access devices **130** is substantially surrounded by the dielectric material **140** that is, in turn, substantially surrounded by the conductive structure **132**. In some such embodiments, the access devices **130** may individually comprise so-called “gate all around” access devices (e.g., gate all around transistors) since each of the access devices **130** is individually substantially surrounded by one of the conductive structures **132**.

(34) The dielectric material **140** may be formed of and include insulative material. By way of non-limiting example, the dielectric material **140** may comprise one or more of phosphosilicate glass, borosilicate glass, borophosphosilicate glass (BPSG), fluorosilicate glass, silicon dioxide, titanium dioxide, zirconium dioxide, hafnium dioxide, tantalum oxide, magnesium oxide, aluminum oxide, niobium oxide, molybdenum oxide, strontium oxide, barium oxide, yttrium oxide, a nitride material, (e.g., silicon nitride ($Si_3N_4$)), an oxynitride (e.g., silicon oxynitride, another gate dielectric material), a dielectric carbon nitride material (e.g., silicon carbon nitride (SiCN)), or a dielectric carboxynitride material (e.g., silicon carboxynitride (SiOCN)).

(35) With continued reference to FIG. **1**C, in some embodiments, the dielectric material **140** may also be located on surfaces of the conductive structures **132** and between the conductive structures **132** and the second insulative material **119**. Portions of the dielectric material **140** on surfaces of the second insulative material **119** may not be referred to as a “gate dielectric” material.

(36) With collective reference to FIG. **1**A and FIG. **1**C, in some embodiments, the conductive structures **132** horizontally terminate (e.g., in the X-direction) at steps **175** of staircase structures **174** formed at horizontal ends (e.g., in the X-direction) of the conductive structures **132** and within a third insulative material **139**. The steps **175** may each individually be at least partially defined by horizontal (e.g., in the X-direction) edges of the conductive structures **132**.

(37) Vertically higher (e.g., in the Z-direction) conductive structures **132** may have a smaller horizontal dimension (e.g., in the X-direction) than vertically lower conductive structures **132**, such that horizontal edges of the conductive structures **132** at least partially define the steps **175** of the staircase structures **174**.

(38) With reference to FIG. **1**A, in some embodiments, the staircase structures **174** are horizontally aligned in a first direction (e.g., in the X-direction) and horizontally offset in a second direction (e.g., the Y-direction). In some embodiments, each vertical stack structure **135** of conductive structures **132** individually includes a staircase structure **174** at a first horizontal end (e.g., in the X-direction) thereof and an additional staircase structure **174** at a second, opposite horizontal end (e.g., in the X-direction) thereof.

(39) In other embodiments, the staircase structures **174** of horizontally neighboring (e.g., in the Y-direction) vertical stack structures **135** of conductive structures **132** may be located at opposing

horizontal ends (e.g., in the X-direction) of the first microelectronic device structure **100**. In some such embodiments, every vertical stack structure **135** of conductive structures **132** (e.g., in the Y-direction) includes a staircase structure **174** at a first horizontal end (e.g., in the X-direction) of the first microelectronic device structure **100** while the other vertical stack structures **135** of conductive structures **132** individually includes a staircase structure **174** at a second horizontal end (e.g., in the X-direction) of the first microelectronic device structure **100** opposite the first horizontal end. Stated another way, the staircase structures **174** of horizontally neighboring (e.g., in the Y-direction) conductive structures **132** may alternate between a first horizontal end (e.g., in the X-direction) of the first microelectronic device structure **100** and a second horizontal end (e.g., in the X-direction) of the first microelectronic device structure **100**, the second horizontal end opposing the first horizontal end.
(40) Although FIG. **1**A illustrates two staircase structures **174** for every vertical stack structure **135** of conductive structures **132** (e.g., a staircase structure **174** at each horizontal end (e.g., in the X-direction) of each vertical stack structure **135** of conductive structures **132**), the disclosure is not so limited. In other embodiments, each vertical stack structure **135** of conductive structures **132** may include one staircase structure **174**, and each of the staircase structures **174** may be located at a same horizontal end (e.g., in the X-direction) of the vertical stack structure **135** of conductive structures **132**.
(41) The quantity of the steps **175** of the staircase structures **174** may correspond to the quantity of the levels of memory cells (e.g., memory cells **120** (FIG. **1**J through FIG. **1**L)) of vertical stacks of the memory cells to be formed in the first microelectronic device structure **100**. Although FIG. **1**A and FIG. **1**C illustrate that the staircase structures **174** individually comprise a particular number (e.g., four (4)) steps **175**, the disclosure is not so limited. In other embodiments, the staircase structures **174** each individually include a desired quantity of the steps **175**, such as within a range from thirty-two (32) of the steps **175** to two hundred fifty-six (256) of the steps **175**. In some embodiments, the staircase structures **174** each individually include sixty-four (64) of the steps **175**. In other embodiments, the staircase structures **174** each individually include ninety-six (96) or more of the steps **175**. In other embodiments, the staircase structures **174** each individually include a different number of the steps **175**, such as less than sixty-four (64) of the steps **175** (e.g., less than or equal to sixty (60) of the steps **175**, less than or equal to fifty (50) of the steps **175**, less than about forty (40) of the steps **175**, less than or equal to thirty (30) of the steps **175**, less than or equal to twenty (20) of the steps **175**, less than or equal to ten (10) of the steps **175**); or greater than sixty-four (64) of the steps **175** (e.g., greater than or equal to seventy (70) of the steps **175**, greater than or equal to one hundred (100) of the steps **175**, greater than or equal to about one hundred twenty-eight (128) of the steps **175**, greater than two hundred fifty-six (256) of the steps **175**).
(42) In some embodiments, the staircase structures **174** each individually include the same quantity of the steps **175**. In some such embodiments, staircase structures **174** of the same vertical stack structure **135** include the same quantity of the steps **175**. In some embodiments, each step **175** of each staircase structure **174** may be vertically offset (e.g., in the Z-direction) from a vertically neighboring step **175** of the staircase structure **174** by one level (e.g., one tier) of the vertically alternating conductive structures **132** and the vertically intervening (e.g., in the Z-direction) dielectric material **140** and second insulative material **119**. In some such embodiments, every conductive structure **132** of the vertical stack structure **135** may comprise a step **175** at each horizontal end (e.g., in the X-direction) of the staircase structures **174** of the vertical stack structure **135**. In other embodiments, vertically neighboring (e.g., in the Z-direction) steps **175** of a staircase structure **174** on a first horizontal size (e.g., in the X-direction) of a vertical stack structure **135** may be vertically offset (e.g., in the Z-direction) by two levels (e.g., two tiers) of the vertically alternating conductive structures **132** and the vertically intervening dielectric material **140** and second insulative material **119**. In some such embodiments, the steps **175** of each staircase structure **174** are formed of every other conductive structure **132** of the vertical stack structure **135** and the

steps **175** of staircase structures **174** at horizontally opposing ends (e.g., in the X-direction) of the same vertical stack structure **135** may be defined by conductive structures **132** that are vertically spaced (e.g., in the Z-direction) from one another by one level of a conductive structure **132** and the vertically intervening dielectric material **140** and second insulative material **119**.

(43) With continued reference to FIG. **1**A and FIG. **1**C, conductive contact structures **176** may be formed in electrical communication with individual steps **175** of the staircase structures **174** and pad structures **178** may be formed in electrical communication with the conductive contact structures **176**. The conductive contact structures **176** may be in electrical communication with individual conductive structures **132** at the steps **175**. For example, the conductive contact structures **176** may individually physically contact (e.g., land on) portions of upper (e.g., in the Z-direction) surfaces of the conductive structures **132** at least partially defining treads of the steps **175**. In some embodiments, every other step **175** of each staircase structure **174** may be in electrical communication with a conductive contact structure **176**. In some such embodiments, each vertical stack structure **135** of conductive structures **132** includes one staircase structure **174** at each horizontal (e.g., in the X-direction) end thereof and every other step **175** of each staircase structure **174** is individually in contact with a conductive contact structure **176**. Each conductive structure **132** of a first staircase structure **174** at a first horizontal end of the vertical stack structure **135** not in electrical communication with a conductive contact structure **176** may individually be in electrical communication with a conductive contact structure **176** at steps **175** of a second staircase structure **174** at a second, opposite horizontal end of the vertical stack structure **135**. In other embodiments, each step **175** of each staircase structure **174** may be in electrical communication with a conductive contact structure **176** at the horizontal (e.g., in the X-direction) end of the staircase structure **174**.

(44) The conductive contact structures **176** and the pad structures **178** may individually be formed of and include conductive material, such as one or more of the materials described above with reference to the conductive structures **132**. In some embodiments, the conductive contact structures **176** and the pad structures **178** comprise substantially the same material composition as the conductive structures **132**. In other embodiments, the conductive contact structures **176** and the pad structures **178** comprise a different material composition than the conductive structures **132**. In some embodiments, the conductive contact structures **176** and the pad structures **178** individually comprise tungsten.

(45) The third insulative material **139** may be formed of and include insulative material, such as one or more of the materials described above with reference to the first insulative material **114**. In some embodiments, the third insulative material **139** is formed of and includes silicon dioxide.

(46) With reference back to FIG. **1**B, vertically neighboring (e.g., in the Z-direction) access devices **130** are spaced from one another by a fourth insulative material **137**. In some embodiments, the fourth insulative material **137** surrounds at least a portion of the dielectric material **140** and horizontally intervenes (e.g., in the Y-direction) between the dielectric material **140** and the second material **118**.

(47) The fourth insulative material **137** may be formed of and include insulative material having an etch selectivity with respect to the second material **118**. In some embodiments, the fourth insulative material **137** comprises a nitride material (e.g., silicon nitride (Si.sub.3N.sub.4)) or an oxynitride material (e.g., silicon oxynitride). In some embodiments, the fourth insulative material **137** comprises silicon nitride.

(48) Conductive pillar structures **160** may vertically extend (e.g., in the Z-direction) through openings **159** within the first microelectronic device structure **100**. The conductive pillar structures **160** may also be referred to herein as “digit lines,” “second conductive lines,” “digit line pillar structures,” “local digit lines,” or “vertical digit lines.” Each conductive pillar structure **160** vertically extends through the first microelectronic device structure **100**, such as through or horizontally neighboring (e.g., in the Y-direction) the vertical stack of access devices **130**. In some

embodiments, the conductive pillar structures **160** horizontally neighbor (e.g., in the Y-direction) a source region or a drain region of the access devices **130**. In other embodiments, such as where the access devices **130** consist essentially of the channel regions **134** (and do not include, for example, a source region and a drain region), the conductive pillar structures **160** directly contact the channel region **134** of the access devices **130**. The conductive pillar structures **160** are individually in electrical communication with the access devices **130** of the vertical stack of access devices **130**.

(49) The conductive pillar structures **160** may individually be formed of and include conductive material, such as one or more of a metal (e.g., one or more of tungsten, titanium, nickel, platinum, rhodium, ruthenium, aluminum, copper, molybdenum, iridium, silver, gold), a metal alloy, a metal-containing material (e.g., metal nitrides, metal silicides, metal carbides, metal oxides), a material including at least one of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), titanium aluminum nitride (TiAlN), iridium oxide ($IrO_x$), ruthenium oxide ($RuO_x$), alloys thereof, a conductively doped semiconductor material (e.g., conductively doped silicon, conductively doped germanium, conductively doped silicon germanium, etc.), polysilicon, or other materials exhibiting electrical conductivity. In some embodiments, the conductive pillar structures **160** comprise tungsten.

(50) As described in further detail herein, the conductive structures **132** may be configured to provide sufficient voltage to a channel region **134** of each of the access devices **130** to electrically couple a storage device (e.g., storage device **150** (FIG. **1**J through FIG. **1**L)) horizontally neighboring (e.g., in the Y-direction) and associated with the access device **130** to, for example, a conductive pillar structure **160** vertically extending (e.g., in the Z-direction) through the first microelectronic device structure **100** and in electrical communication with the vertical stack of access devices **130**. Stated another way, each conductive structure **132** may individually comprise a gate structure configured to provide a sufficient voltage to the channel region **134** vertically neighboring (e.g., in the Z-direction) the conductive structure **132** to electrically couple the access device **130** to a horizontally neighboring (e.g., in the Y-direction) storage device.

(51) In some embodiments, horizontally neighboring (e.g., in the X-direction, in the Y-direction) conductive pillar structures **160** within an opening **159** are in electrical communication with one another through conductive material **163** horizontally extending between and electrically connecting the neighboring conductive pillar structures **160**. The conductive material **163** may be formed of and include conductive material, such as one or more of the materials described above with reference to the conductive pillar structures **160**. In some embodiments the conductive material **163** comprises substantially the same material composition as the conductive pillar structures **160**. As described in further detail herein, the horizontally extending conductive material **163** may be removed to electrically isolate the conductive pillar structures **160** from one another to form vertical stacks of memory cells, each of the conductive pillar structures **160** individually in electrical communication with a vertical stack of access devices **130** of a different vertical stack of memory cells (e.g., memory cells **120** (FIG. **1**J through FIG. **1**L)).

(52) A dimension $D_1$ (e.g., a diameter) of the openings **159** between opposing surfaces of the conductive pillar structures **160** may be within a range of from about 60 nanometers (nm) to about 120 nm, such as from about 60 nm to about 80 nm, from about 80 nm to about 100 nm, or from about 100 nm to about 120 nm. In some embodiments, the dimension $D_1$ is from about 80 nm to about 100 nm.

(53) In some embodiments, a distance $D_2$ between a surface of the conductive pillar structure **160** and the conductive structures **132** (corresponding to the horizontal length (e.g., in the Y-direction) of the access device **130** between the conductive pillar structure **160** and the conductive structure **132**) may be within a range of from about 10 nm to about 50 nm, such as from about 10 nm to about 20 nm, from about 20 nm to about 30 nm, from about 30 nm to about 40 nm, or from about 40 nm to about 50 nm. In some embodiments, the distance $D_2$ is about 5 nm. However, the distance $D_2$ may be different than (e.g., less than, greater than) those described above. The

distance D.sub.2 may be referred to herein as a “junction” length between the conductive pillar structure **160** and the access device **130**. In some embodiments, the distance D.sub.2 corresponds to a horizontal length (e.g., in the Y-direction) of the channel region **134** between the conductive pillar structure **160** and the conductive structure **132**.

(54) With reference to FIG. **1**B and FIG. **1**C, a semiconductive material **170** vertically overlies (e.g., in the Z-direction) a vertically (e.g., in the Z-direction) uppermost level of the first insulative material **114**. In some embodiments, the semiconductive material **170** comprises silicon. In some embodiments, the semiconductive material **170** comprises single-crystal silicon. As described in further detail herein, the semiconductive material **170** may be used to form one or more control logic devices of the first microelectronic device structure **100** to facilitate control operations for memory cells (e.g., memory cells **120** (FIG. **1**J through FIG. **1**L)) of the first microelectronic device structure **100**.

(55) A fifth insulative material **161** vertically overlies (e.g., in the Z-direction) the semiconductive material **170**. The fifth insulative material **161** may be formed of and include one or more insulative materials, such as one or more of the materials described above with reference to the first insulative material **114**. In some embodiments, the fifth insulative material **161** comprises silicon dioxide.

(56) With reference to FIG. **1**A, although the fifth insulative material **161** vertically overlies (e.g., in the Z-direction) portions of the first microelectronic device structure **100**, the fifth insulative material **161** is not illustrated in FIG. **1**A. Rather, for clarity and ease of understanding the description, only portions of the first microelectronic device structure **100** are illustrated in FIG. **1**A to illustrate the relative position of different portions of the first microelectronic device structure **100**.

(57) FIG. **1**D is a simplified partial cross-sectional view of the first microelectronic device structure **100** at a later processing stage than that illustrated in FIG. **1**A through FIG. **1**C. The cross-sectional view of FIG. **1**D illustrates the same cross-sectional view illustrated in FIG. **1**B.

(58) With reference to FIG. **1**D, an insulative liner material **190** may be formed within the openings **159** and vertically over the first microelectronic device structure **100**. The insulative liner material **190** may overlie and contact portions of the conductive pillar structures **160** within the openings **159**.

(59) In some embodiments, the insulative liner material **190** is formed to a thickness T.sub.1 such that the insulative liner material **190** does not completely fill the openings **159**. In some such embodiments, the thickness T.sub.1 may be less than one-half the dimension D.sub.1 of the openings **159**. In some embodiments, the thickness T.sub.1 is less than about one-third, such as less than about one-quarter of the dimension D.sub.1 of the openings **159**.

(60) The thickness T.sub.1 may be within a range of from about 10 nm to about 30 nm, such as from about 10 nm to about 20 nm, or from about 20 nm to about 30 nm. In some embodiments, the thickness T.sub.1 is about 20 nm. In some embodiments, a distance D.sub.3 between opposing portions of the insulative liner material **190** within the openings **159** may be within a range of from about 30 nm to about 60 nm, such as from about 30 nm to about 40 nm, from about 40 nm to about 50 nm, or from about 50 nm to about 60 nm. In some embodiments, the distance D.sub.3 is at least about 30 nm. In other embodiments, the distance D.sub.3 is at least about 40 nm.

(61) In some embodiments, the thickness T.sub.1 of the insulative liner material **190** is less than about thirty percent (30%) of the dimension D.sub.1 (FIG. **1**B) such that at least about forty percent of the dimension D.sub.1 of the opening **159** (FIG. **1**B) remains open to facilitate dry etching of vertically lower (e.g., in the Z-direction) portions of the openings **159** to remove material (e.g., portions of the insulative liner material **190**, portions of the conductive pillar structures **160**) from vertically lower portions of the openings **159**.

(62) The insulative liner material **190** may comprise a material configured and formulated such that sidewalls of the conductive pillar structures **160** are not substantially oxidized by formation of

(e.g., deposition of, such as by ALD) the insulative liner material **190**. In some embodiments, exposed surface of the conductive pillar structures **160** are exposed to hydrogen (H.sub.2) gas prior to formation of the insulative liner material **190**.

(63) The insulative liner material **190** may be formed of and include insulative material. By way of non-limiting example, in some embodiments, the insulative liner material **190** is formed of and includes an oxide material (e.g., silicon dioxide, zirconium oxide, hafnium oxide, aluminum oxide), a nitride material (e.g., silicon nitride), or an oxynitride (e.g., silicon oxynitride (SiON)). In some embodiments, the insulative liner material **190** comprises an oxide material.

(64) FIG. **1**E is a simplified partial cross-sectional view of the first microelectronic device structure **100** at a later processing stage than that illustrated in FIG. **1**D and illustrating the same cross-sectional view as that illustrated in FIG. **1**D.

(65) With reference to FIG. **1**E, vertically lower (e.g., in the Z-direction) portions of the insulative liner material **190** may be removed through the openings **159** (FIG. **1**D) to expose the conductive material **163** horizontally extending (e.g., in the X-direction, in the Y-direction) between horizontally neighboring conductive pillar structures **160**. In some embodiments, horizontally extending (e.g., in the X-direction, in the Y-direction) portions of the insulative liner material **190** are removed, such as by exposing the insulative liner material **190** to one or more dry etchants through the openings **159**. The one or more dry etchants may be formulated and configured to substantially remove the horizontally extending portions of the insulative liner material **190** without substantially removing the conductive pillar structures **160**. Stated another way, the one or more dry etchants may be selective to the insulative liner material **190** to selectively remove portions of the insulative liner material **190** relative to the conductive pillar structures **160**.

(66) In some embodiments, exposing the insulative liner material **190** to one or more dry etchants and removing the horizontally extending portions of the insulative liner material **190** from vertically lower (e.g., in the Z-direction) portions of the openings **159** (FIG. **1**D) includes horizontally thinning (e.g., in the X-direction, in the Y-direction) vertically extending (e.g., in the Z-direction) portions of the insulative liner material **190** through the openings **159**. In some embodiments, a thickness T.sub.2 of the insulative liner material **190** after removal of the horizontally extending portions of the insulative liner material **190** may be less than the thickness T.sub.1 (FIG. **1**D) of the insulative liner material **190** prior to removal of the horizontally extending portions of the insulative liner material **190**. The thickness T.sub.2 may be within a range of from about 3 nm to about 15 nm, such as from about 3 nm to about 5 nm, from about 5 nm to about 10 nm, or from about 10 nm to about 15 nm. In some embodiments, the thickness T.sub.2 is about 5 nm.

(67) A distance D.sub.4 between opposing portions of the insulative liner material **190** may be within a range of from about 40 nm to about 70 nm, such as from about 40 nm to about 50 nm, from about 50 nm to about 60 nm, or from about 60 nm to about 70 nm. However, the disclosure is not so limited and the distance D.sub.4 may be different than those described above.

(68) With continued reference to FIG. **1**E, after removing the horizontally extending (e.g., in the X-direction, in the Y-direction) portions of the insulative liner material **190**, the horizontally extending conductive material **163** electrically connecting horizontally neighboring conductive pillar structures **160** within the openings **159** may be removed. In some embodiments, the conductive material **163** is exposed to an etchant formulated and configured to selectively remove the exposed portions of the conductive material **163** and expose the vertically underlying (e.g., in the Z-direction) first insulative material **114**.

(69) The conductive material **163** at the vertically lower (e.g., in the Z-direction) portion of the openings **159** (FIG. **1**D) may be removed by exposing the conductive material **163** to one or more wet etchants. By way of non-limiting example, the conductive material **163** may be exposed to one or more of ammonium hydroxide (NH.sub.4OH), hydrogen peroxide (H.sub.2O.sub.2), deionized (DI) water, hydrofluoric acid (HF), and nitric acid (HNO.sub.3).

(70) Removing the conductive material **163** may form isolated conductive pillar structures **160**. For example, each opening **159** (FIG. **1**D) may include two isolated conductive pillar structures **160**, each of which is individually in electrical communication with the horizontally neighboring vertical stack of access devices **130**. As described in further detail herein, each isolated conductive pillar structure **160** forms a portion of a vertical stack of memory cells (e.g., memory cells **120** (FIG. **1**J through FIG. **1**L)) and is configured to be in electrical communication with storage devices (e.g., storage devices **150** (FIG. **1**J through FIG. **1**L)) of the vertical stack of memory cells by means of the access devices **130**.
(71) A sixth insulative material **191** may be formed within the openings **159** (FIG. **1**D) and vertically over (e.g., in the Z-direction) exposed portions of the insulative liner material **190** defining sidewalls of the openings **159** and on the exposed portions of the first insulative material **114** at the vertically lowermost (e.g., in the Z-direction) portion of the openings **159**.
(72) The sixth insulative material **191** may be formed of and include insulative material, such as one or more of the materials described above with reference to the first insulative material **114**. In some embodiments, the sixth insulative material **191** comprises silicon dioxide.
(73) FIG. **1**F is a simplified partial cross-sectional view of the first microelectronic device structure **100** at a later processing stage than that illustrated in FIG. **1**E and illustrating the same cross-sectional view as that illustrated in FIG. **1**E.
(74) With reference to FIG. **1**F, the first microelectronic device structure **100** may be further processed to form vertical stacks of memory cells (e.g., memory cells **120** (FIG. **1**J through FIG. **1**L)) including vertical stacks of storage devices (e.g., storage devices **150** (FIG. **1**J through FIG. **1**L)) configured to be in electrical communication with the vertical stacks of access devices **130**.
(75) In some embodiments, portions of the sixth insulative material **191** and the fifth insulative material **161** (FIG. **1**E) may be removed, such as by exposing the first microelectronic device structure **100** to a chemical mechanical planarization (CMP) process. In some embodiments, removing the sixth insulative material **191** and the fifth insulative material **161** exposes a surface of the semiconductive material **170**.
(76) Trenches **165** may be formed in regions horizontally between (e.g., in the Y-direction) horizontally neighboring (e.g., in the Y-direction) conductive pillar structures **160** of neighboring openings **159** (FIG. **1**D). For example, trenches **165** may be formed through the first microelectronic device structure **100** horizontally between (e.g., in the Y-direction) access devices **130** that are located horizontally between (e.g., in the Y-direction) conductive pillar structures **160**. The trenches **165** may be formed vertically through (e.g., in the Z-direction) the semiconductive material **170** and the levels of the first material **116** and the second material **118**. In some embodiments, the trenches **165** expose a portion of the first insulative material **114**.
(77) After forming the trenches **165**, portions of the first material **116** may be removed selective to the second material **118** through the trenches **165**. In some embodiments, the first material **116** is selectively removed by exposing the first material **116** to one or both of a dry etch process (e.g., with one or more of sulfur hexafluoride (SF.sub.6), hydrogen (H.sub.2), and carbon tetrafluoride (CF.sub.4)) or a wet etch process (e.g., with one or more quaternary ammonium compounds (e.g., one or more of benzyltrimethyl ammonium hydroxide (C.sub.10H.sub.17NO), methyltriethylammonium hydroxide (C.sub.7H.sub.19NO), ethyltrimethyl ammonium hydroxide (ETMAH) (C.sub.5H.sub.15NO), 2-hydroxyethyltrimethyl ammonium hydroxide (also referred to as “choline hydroxide”) (C.sub.5H1.sub.5NO.sub.2), benzyltriethyl ammonium hydroxide (C.sub.10H.sub.17NO), hexadecyltrimethyl ammonium hydroxide (C.sub.19H.sub.43NO)) and one or more amine compounds (e.g., one or more of N-methylethanolamine (NMEA) (C.sub.3H.sub.9NO), monoethanolamine (MEA) (C.sub.2H.sub.7NO), diethanolamine (DEA) (C.sub.4H.sub.11NO.sub.2), triethanolamine (TEA) (C.sub.6H.sub.15NO.sub.3), triisopropanolamine (C.sub.9H.sub.21NO.sub.3), 2-(2-aminoethylamino)ethanol (C.sub.4H.sub.12N.sub.2O), 2-(2-aminoethoxy)ethanol (AEE) (C.sub.4H.sub.11NO.sub.2), N-

ethyl ethanolamine ($C_4H_{11}NO$), N,N-dimethylethanolamine ($C_4H_{11}NO$), N,N-diethyl ethanolamine ($C_6H_{15}NO$), N-methyl diethanolamine (MDEA) ($C_5H_{13}NO_2$), N-ethyl diethanolamine ($C_6H_{15}NO_2$), cyclohexylaminediethanol ($C_{10}H_{21}N$), diisopropanolamine ($C_6H_{15}NO_2$), cyclohexylaminediethanol) ($C_{10}H_{21}N$)). However, the disclosure is not so limited and the first material **116** may be selectively removed with materials and methods other than those described above. In some embodiments, removal of the first material **116** may remove portions of the second material **118** to form voids having a larger vertical dimension (e.g., in the Y-direction) than the first material **116**.

(78) After removal of the portions of the first material **116** and the second material **118**, a first electrode material **152** may be formed on exposed surfaces within the trenches **165**, such as on exposed surfaces of the second material **118**, the access devices **130**, and a portion of the fourth insulative material **137**. In some embodiments, the first electrode material **152** is formed by deposition, such as by ALD.

(79) The first electrode material **152** may be formed of and include conductive material such as, for example, one or more of a metal (e.g., tungsten, titanium, nickel, platinum, rhodium, ruthenium, aluminum, copper, molybdenum, iridium, silver, gold), a metal alloy, a metal-containing material (e.g., metal nitrides, metal silicides, metal carbides, metal oxides), a material including at least one of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), titanium aluminum nitride (TiAlN), iridium oxide ($IrO_x$), ruthenium oxide ($RuO_x$), alloys thereof, a conductively doped semiconductor material (e.g., conductively doped silicon, conductively doped germanium, conductively doped silicon germanium), polysilicon, and other materials exhibiting electrical conductivity. In some embodiments, the first electrode material **152** comprises titanium nitride.

(80) FIG. **1**G is a simplified partial cross-sectional view of the first microelectronic device structure **100** at a processing stage after that illustrated in FIG. **1**F. FIG. **1**G illustrates the same cross-section as that illustrated in FIG. **1**F. With reference to FIG. **1**G, after forming the first electrode material **152** in the trenches **165**, portions of the first electrode material **152** may be removed. For example, vertically extending (e.g., in the Z-direction) portions of the first electrode material **152** on surfaces of the semiconductive material **170** and of the first insulative material **114** may selectively be removed.

(81) After removing the portions of the first electrode material **152**, a dielectric material **156** may be formed on exposed surfaces within the trenches **165**, such as on exposed surfaces of the first insulative material **114**, the first electrode material **152**, the fourth insulative material **137**, and the semiconductive material **170**. In some embodiments, the dielectric material **156** is formed by deposition, such as by ALD.

(82) The dielectric material **156** may be formed of and include one or more of silicon dioxide ($SiO_2$), silicon nitride ($Si_3N_4$), polyimide, titanium dioxide ($TiO_2$), tantalum oxide ($Ta_2O_5$), aluminum oxide ($Al_2O_3$), an oxide-nitride-oxide material (e.g., silicon dioxide-silicon nitride-silicon dioxide), strontium titanate ($SrTiO_3$) (STO), barium titanate ($BaTiO_3$), hafnium oxide ($HfO_2$), zirconium oxide ($ZrO_2$), a ferroelectric material (e.g., ferroelectric hafnium oxide, ferroelectric zirconium oxide, lead zirconate titanate (PZT)), and a high-k dielectric material.

(83) With reference to FIG. **1**H, which illustrates the same cross-section as FIG. **1**G at a later processing stage than that illustrated in FIG. **1**G, after forming the dielectric material **156**, a second electrode material **154** may be formed on surfaces of the dielectric material **156** within the trenches **165** to form storage devices **150**, each individually comprising the first electrode material **152**, the second electrode material **154**, and the dielectric material **156** between the first electrode material **152** and the second electrode material **154**. Accordingly, each of the storage devices **150** individually comprises a first electrode material **152** (also referred to herein as “a first electrode,” “an outer electrode,” “a first electrode plate,” or a “first node structure”), a second electrode

material **154** (also referred to herein as “a second electrode,” “an inner electrode,” “a second electrode plate,” or a “second node structure”), and a dielectric material **156** between the first electrode material **152** and the second electrode material **154**. In some such embodiments, the storage devices **150** individually comprise capacitors. However, the disclosure is not so limited and in other embodiments, the storage devices **150** may each individually comprise other structures, such as, for example, phase change memory (PCM), resistance random-access memory (RRAM), conductive-bridging random-access memory (conductive bridging RAM), or another structure for storing a logic state.

(84) In some embodiments, each access device **130** of the vertical stacks of access devices **130** is horizontally neighbored (e.g., in the Y-direction) by a storage device **150** of a corresponding vertical stack of storage devices **150** to form a vertical stack of memory cells **120**. Each memory cell **120** comprises one of the storage devices **150** in electrical communication with a horizontally neighboring access device **130**. In some embodiments, each memory cell **120** comprises a dynamic random access memory (DRAM) cell. Each memory cell **120** individually comprises a storage device **150** horizontally neighboring an access device **130** of the same level. Accordingly, the vertical stack of memory cells **120** comprises vertically neighboring (e.g., in the Z-direction) levels of memory cells **120**, each level of memory cells **120** comprising an access device **130** and a horizontally neighboring storage device **150**. In other words, each vertical stack of memory cells **120** comprises vertically spaced (e.g., in the Z-direction) levels of memory cells **120**, each vertical level of each vertical stack of memory cells **120** comprising a vertical level of a vertical stack of access devices **130** and a vertical level of a vertical stack of storage devices **150**. Stated another way, each vertical stack of memory cells **120** comprises a vertical stack of access devices **130** and a vertical stack of storage devices **150**, the storage devices **150** of the vertical stack of storage devices **150** coupled to the access devices **130** of the vertical stack of access devices **130**. The vertical stack of access devices **130** may horizontally neighbor (e.g., in the X-direction) the vertical stack of storage devices **150**.

(85) The second electrode material **154** may be formed by deposition, such as by ALD. The second electrode material **154** may be formed of and include conductive material, such as one or more of the materials described above with reference to the first electrode material **152**. In some embodiments, the second electrode material **154** comprises one or more of the materials described above with reference to the first electrode material **152**. In some embodiments, the second electrode material **154** comprises substantially the same material composition as the first electrode material **152**.

(86) FIG. **1**I is a simplified partial cross-sectional view of the first microelectronic device structure **100** at a processing stage after that illustrated in FIG. **1**H and illustrates the same cross-section as that illustrated in FIG. **1**H. With reference to FIG. **1**I, after forming the second electrode material **154** and forming the memory cells **120** comprising the storage devices **150** and the access devices **130**, a conductive plate structure **142** may be formed within remaining portions of the trenches **165** (FIG. **1**H) and on surfaces of the second electrode material **154**.

(87) The conductive plate structures **142** may individually be in electrical communication with the second electrode materials **154** of storage devices **150** of horizontally neighboring vertical stacks of storage devices **150** of horizontally neighboring vertical stacks of memory cells **120**. Accordingly, each of the storage devices **150** of the vertical stack of storage devices **150** may be in electrical communication with a conductive plate structure **142** vertically extending (e.g., in the Z-direction) through the first microelectronic device structure **100**.

(88) The second electrode material **154** may be in electrical communication with one of the conductive plate structures **142** of a vertical stack of memory cells **120**. In some embodiments, the second electrode materials **154** are substantially integral with the conductive plate structures **142**. In some embodiments, the second electrode materials **154** of horizontally neighboring (e.g., in the Y-direction) vertical stacks of storage devices **150** are in electrical communication with the same

conductive plate structure **142**. In some embodiments, and with reference to FIG. **1**J, the second electrode materials **154** of horizontally neighboring (e.g., in the X-direction) vertical stacks of storage devices **150** that directly horizontally neighbor (e.g., in the X-direction) one another and are not separated by, for example, an access device **130**, are in electrical communication with the same conductive plate structure **142**.
(89) In some embodiments, the conductive plate structures **142** are individually formed of conductive material, such as one or more of the materials of the second electrode material **154**. In some embodiments, the conductive plate structures **142** comprise substantially the same material composition as the second electrode material **154**. In some such embodiments, the conductive plate structures **142** are formed substantially concurrently with formation of the second electrode material **154**, which may be formed by, for example, CVD. In other embodiments, the conductive plate structures **142** comprise a different material composition than the second electrode material **154**.
(90) The conductive plate structures **142** may be referred to herein as “conductive plates” or “ground structures.” With reference to FIG. **1**J, in some embodiments, the conductive plate structures **142** horizontally extend (e.g., in the X-direction) as conductive plates. In some embodiments, the conductive plate structures **142** horizontally extend in substantially the same direction and are substantially parallel to the conductive structures **132**. Referring to FIG. **1**J, the conductive plate structures **142** may be horizontally between (e.g., in the Y-direction) vertical stacks of memory cells **120**, such as between vertical stacks of storage devices **150**.
(91) FIG. **1**J through FIG. **1**L are a simplified partial top down view (FIG. **1**J) and simplified partial cross-sectional views (FIG. **1**K, FIG. **1**L) of the first microelectronic device structure **100** at a processing stage after that illustrated in FIG. **1**I. FIG. **1**K is a simplified partial cross-sectional view of the first microelectronic device structure **100** taken through section line K-K of FIG. **1**J; and FIG. **1**L is a simplified partial cross-sectional view of the first microelectronic device structure **100** taken through section line L-L of FIG. **1**J.
(92) With reference to FIG. **1**K and FIG. **1**L, isolation trenches **186** may be formed within the semiconductive material **170** to electrically isolate different portions of the semiconductive material **170** to facilitate formation of transistor structures **185** within a seventh insulative material **187**.
(93) The transistor structures **185** may each include conductively doped regions **188**, each of which includes a source region **188**A and a drain region **188**B. Channel regions of the transistor structures **185** may be horizontally interposed between the conductively doped regions **188**. In some embodiments, the conductively doped regions **188** of each transistor structure **185** individually comprises one or more semiconductive materials doped with at least one conductivity enhancing chemical species, such as at least one N-type dopant (e.g., one or more of arsenic, phosphorous, antimony, and bismuth) or at least one P-type dopant (e.g., one or more of boron, aluminum, and gallium). In some embodiments, the conductively doped regions **188** comprise conductively doped silicon.
(94) The transistor structures **185** include multiplexers **185**A and additional transistors comprising so-called “bleeder” transistors **185**B (also referred to as “leaker” transistors). The multiplexers **185**A horizontally neighbor (e.g., in the Y-direction) the conductive pillar structures **160** and the bleeder transistors **185**B horizontally neighbor (e.g., in the Y-direction) the conductive plate structures **142**. The multiplexers **185**A and the bleeder transistors **185**B each individually comprise a gate structure **189** vertically overlying the semiconductive material **170** and horizontally extending between conductively doped regions **188**. The gate structure **189** of the multiplexers **185**A comprises multiplexer gates **189**A and the gate structure **189** of the bleeder transistors **185**B comprises bleeder gates **189**B. One of the multiplexers **185**A is illustrated in box **185**A in FIG. **1**J and one of the bleeder transistors **185**B is illustrated in box **185**B in FIG. **1**J.
(95) In FIG. **1**K, the source regions **188**A are illustrated in broken lines to indicate that the source

regions **188**A are located in a different plane than that of FIG. **1**K. With reference to FIG. **1**L, the drain regions **188**B are illustrated in broken lines to indicate that they are located in a different plane than that illustrated in FIG. **1**L. With reference to FIG. **1**J, the source regions **188**A are horizontally offset (e.g., in the X-direction, in the Y-direction) from the drain regions **188**B. With collective reference to FIG. **1**J and FIG. **1**K, in some embodiments, the drain regions **188**B of the multiplexers **185**A and the drain regions **188**B of the bleeder transistors **185**B are horizontally aligned (e.g., in the X-direction), such as with a vertically overlying (e.g., in the Z-direction) global digit line **108**.

(96) In some embodiments, each multiplexer **185**A horizontally neighbors (e.g., in the X-direction, in the Y-direction) one of the bleeder transistors **185**B. In some embodiments, two bleeder transistors **185**B horizontally intervene (e.g., in the Y-direction) between horizontally neighboring multiplexers **185**A. In some embodiments, a conductive plate structure **142** may be located between horizontally neighboring (e.g., in the Y-direction) conductive pillar structures **160**. Two multiplexers **185**A may horizontally intervene (e.g., in the Y-direction) between the conductive pillar structures **160** and two bleeder transistors **185**B may be horizontally intervene between the multiplexers **185**A. In some such embodiments, the multiplexers **185**A may be located closer to the conductive pillar structures **160** than the bleeder transistors **185**B and the bleeder transistors **185**B may be closed to the conductive plate structure **142** than the multiplexers **185**A.

(97) In some embodiments, the source region **188**A of the multiplexer **185**A is shared with the horizontally neighboring bleeder transistor **185**B. Stated another way, each multiplexer **185**A comprises a source region **188**A that is shared with a source region **188**A of the horizontally neighboring bleeder transistor **185**B. In other words, the source region **188**A of the multiplexer **185**A comprises the source region **188**A of the bleeder transistor **185**B. In some such embodiments, the multiplexers **185**A and the horizontally neighboring bleeder transistor **185**B may be referred to as “shared source” transistors.

(98) In some embodiments, each of the source regions **188**A and the drain regions **188**B are individually in electrical communication with first conductive interconnect structures **192**. The first conductive interconnect structures **192** may individually electrically couple the source regions **188**A and the drain regions **188**B to one or more additional conductive structures, as described in further detail herein.

(99) Each of the multiplexer gates **189**A and the bleeder gates **189**B may individually be horizontally aligned (e.g., in the Y-direction) with and shared by the channel regions of multiple transistor structures **185** (e.g., multiplexers **185**A, bleeder transistors **185**B) horizontally neighboring (e.g., in the X-direction (FIG. **1**A)) one another. In some such embodiments, and as illustrated in FIG. **1**J, the gate structures **189** extend in a first horizontal direction (e.g., in the X-direction). In addition, dielectric material (also referred to herein as a “gate dielectric material”) may be vertically interposed between the gate structures **190** and portions of the semiconductive material **170** at least partially defining the channel regions of the transistor structures **185**.

(100) With reference to FIG. **1**J, the multiplexer gates **189**A horizontally extend (e.g., in the X-direction) and are in electrical communication with a multiplexer driver **197**. The multiplexer driver **197** may include circuitry configured to drive the multiplexers **185**A by means of the multiplexer gates **189**A. The bleeder gates **189**B may horizontally extend (e.g., in the X-direction) and be in electrical communication with a bleeder driver **199** including circuitry configured to driver the bleeder transistors **185**B by means of the bleeder gates **189**B. In some embodiments, each of the multiplexer drivers **197** and the bleeder drivers **199** are located in the peripheral region **103** of the first microelectronic device structure **100**. In some embodiments, the multiplexer drivers **197** and the bleeder drivers **199** are located within horizontal boundaries (e.g., in the X-direction) of the staircase structures **174**.

(101) With reference to FIG. **1**K, global digit line contact structures **162** may be formed in electrical communication with the multiplexers **185**A by means of first routing structures **194** in

electrical communication with a first conductive interconnect structure **192** that is, in turn, in electrical communication with a drain region **188**B of the multiplexer **185**A. The global digit line contact structures **162** are individually in electrical communication with one of the global digit lines **108**. In some embodiments, each global digit line **108** is configured to be in electrical communication with more than one vertical stacks of memory cells **120** (e.g., more than one conductive pillar structure (local digit line) **160** of different vertical stacks of memory cells **120**). In some such embodiments, the global digit lines **108** are individually configured to be in electrical communication with more than one of the global digit line contact structures **162**.

(102) The global digit lines **108** may also be referred to as “conductive lines.” The global digit lines **108** may be formed vertically over (e.g., in the Z-direction) and in electrical communication with the global digit line contact structures **162**. With collective reference to FIG. **1**J and FIG. **1**K, the global digit lines **108** are located within the array region **101** and extend in a horizontal direction (e.g., the Y-direction) substantially perpendicular to the conductive structures **132**. The global digit lines **108** may terminate at horizontally terminal ends (e.g., in the Y-direction) of the array region **101**.

(103) With continued reference to FIG. **1**K, the global digit lines **108** may vertically overlie (e.g., in the Z-direction) the vertical stacks of memory cells **120** and the semiconductive material **170** including the multiplexers **185**A and the bleeder transistors **185**B. In some embodiments, the multiplexers **185**A and the bleeder transistors **185**B vertically intervene (e.g., in the Z-direction) between the global digit lines **108** and the vertical stacks of memory cells **120** and the vertical stacks of conductive structures **132**.

(104) The global digit lines **108** include first global digit lines **108**A and second global digit lines **108**B. The first global digit lines **108**A may be referred to herein as “through global digit lines” and the second global digit lines **108**B may be referred to herein as “reference global digit lines.” The first global digit lines **108**A and the second global digit lines **108**B may collectively be referred to herein as “global digit lines.” In some embodiments, the first global digit lines **108**A are located on a first horizontal end (e.g., in the Y-direction) of the first microelectronic device structure **100** and the second global digit lines **108**B are located on a second horizontal end (e.g., in the Y-direction) of the first microelectronic device structure **100** opposite the first horizontal end. For example, in the view illustrated in FIG. **1**A, the first global digit lines **108**A may be located in the upper horizontal half (e.g., in the Y-direction) of the array region **101** and the second global digit lines **108**B may be located in a lower horizontal half (e.g., in the Y-direction) of the array region **101**.

(105) Each of the global digit lines **108** and the global digit line contact structures **162** may individually be formed of and include conductive material, such as, for example, one or more of a metal (e.g., tungsten, titanium, nickel, platinum, rhodium, ruthenium, aluminum, copper, molybdenum, iridium, silver, gold), a metal alloy, a metal-containing material (e.g., metal nitrides, metal silicides, metal carbides, metal oxides), a material including at least one of titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), titanium aluminum nitride (TiAlN), iridium oxide ($IrO_x$), ruthenium oxide ($RuO_x$), alloys thereof, a conductively doped semiconductor material (e.g., conductively doped silicon, conductively doped germanium, conductively doped silicon germanium, etc.), polysilicon, or other materials exhibiting electrical conductivity. In some embodiments, the global digit lines **108** and the global digit line contact structures **162** individually comprise tungsten. In other embodiments, the global digit lines **108** and the global digit line contact structures **162** individually comprise copper.

(106) In some embodiments, the drain regions **188**B of the multiplexers **185**A and the bleeder transistors **185**B may be horizontally aligned (e.g., in the X-direction) with the global digit line **108** vertically overlying (e.g., in the Z-direction) the multiplexers **185**A and the bleeder transistors **185**B. In some embodiments, the source regions **188**A may be horizontally offset (e.g., in the X-direction) from the vertically overlying global digit line **108**.

(107) Still referring to FIG. **1**K, the conductive plate structures **142** are individually in electrical

communication with a second conductive interconnect structure **195**. The second conductive interconnect structure **195** is, in turn, in electrical communication with a second routing structure **172**. The second routing structure **172** is in electrical communication with first conductive interconnect structures **192** that are each, in turn, in electrical communication with the drain regions **188**B of the bleeder transistors **185**B. Accordingly, the drain regions **188**B of the bleeder transistors **185**B horizontally neighboring (e.g., in the Y-direction) the conductive plate structures **142** are in electrical communication with the conductive plate structures **142** by means of the first conductive interconnect structures **192**, the second routing structure **172**, and the second conductive interconnect structures **195**. The second routing structures **172** are not illustrated in the top-down view of FIG. **1**J for clarity and ease of understanding the description, but it will be understood that the second routing structures horizontally extend (e.g., in the Y-direction) and are substantially parallel with the global digit lines **108**.

(108) With reference to FIG. **1**L, the shared source regions **188**A may be in electrical communication with a first conductive interconnect structure **192** that is, in turn, in electrical communication with a third routing structure **173**. The third routing structures **173** horizontally extend (e.g., in the Y-direction) between the first conductive interconnect structure **192** in electrical communication with the source region **188**A and a third conductive interconnect structure **196** that is in electrical communication with the conductive pillar structure **160**. Accordingly, in some embodiments, the conductive pillar structures **160** are in electrical communication with the source regions **188**A of the multiplexers **185**A by means of the third conductive interconnect structure **196**, the third routing structure **173**, and the first conductive interconnect structure **192**.

(109) In some embodiments, the multiplexers **185**A may be configured to selectively place the vertical stacks of memory cells **120** in electrical communication with a global digit line **108** by means of a global digit line contact structure **162**. By way of non-limiting example, a select voltage may be applied to the multiplexer gate **189**A to electrically connect the global digit line **108** to a vertical stack of memory cells **120** associated with the multiplexer **185**A comprising the multiplexer gate **189**A. Application of the select voltage to the multiplexer gate **189**A may place the conductive pillar structure **160** of the selected vertical stack of memory cells **120** in electrical communication with the global digit line **108** in electrical communication with the conductive pillar structure **160**. Accordingly, the conductive pillar structure **160** (e.g., the local digit line) of each vertical stack of memory cells **120** may selectively be coupled to the global digit line **108** by means of the multiplexer **185**A between the conductive pillar structure **160** and the global digit line **108**.

(110) Each global digit line **108** may be configured to be selectively coupled to more than one of the conductive pillar structures **160** by means of the multiplexers **185**A coupled to global digit lines **108**. Each of the multiplexers **185**A is configured to be in electrical communication with the global digit line **108** and one of the multiplexers **185**A by means of one of the global digit line contact structures **162**. In some embodiments, each global digit line **108** is configured to selectively be in electrical communication with four (4) of the conductive pillar structures **160**, each one of which is associated with a vertical stack of memory cells **120**. In other embodiments, each of the global digit lines **108** is configured to selectively be in electrical communication with eight (8) of the conductive pillar structures **160** or sixteen (16) of the conductive pillar structures **160**. In use and operation, application of a voltage to the multiplexer gate **189**A induces a current in the channel region of the multiplexer **185**A and electrically connects the global digit line **108** (e.g., by means of the global digit line contact structure **162**, the first routing structure **194**, one of the first conductive interconnect structures **192**, and the drain region **188**B of the multiplexer **185**A) to the conductive pillar structure **160** (e.g., by means of the source region **188**A of the multiplexer **185**A, the first conductive interconnect structure **192** in electrical communication with the source region **188**A, the third routing structure **173**, and the third conductive interconnect structure **196**). Accordingly, in some embodiments, the multiplexers **185**A are individually configured to receive a signal (e.g., a

select signal) from a multiplexer driver **197** and provide the signal to a bit line (e.g., conductive pillar structures **160** (FIG. **1**L)) to selectively access desired memory cells **120** within the array region **101** for effectuating one or more control operations of the memory cells **120**.
(111) The bleeder transistors **185**B may be configured to provide a bias voltage from the conductive plate structure **142** on the source region **188**A of multiplexers **185**A of unselected vertical stacks of memory cells **120**. For example, as described above, a particular vertical stack of memory cells **120** may be selected to be in electrical communication with a global digit line **108** by means of a multiplexer **185**A in electrical communication with the global digit line **108** through a global digit line contact structure **162**. The global digit line **108** may be configured to be in electrical communication with additional vertical stacks of memory cells **120** by means of additional multiplexers **185**A configured to be electrically connect the global digit line **108** with the additional vertical stacks of memory cells **120**. The bleeder transistors **185**B may be configured to place the source regions **188**A of the transistor structures **185** (e.g., the multiplexers **185**A) of unselected vertical stacks of memory cells **120** (e.g., vertical stacks of memory cells **120** not in electrical communication with the global digit line **108**, such as memory cells **120** of a vertical stack in which a select voltage is not applied to the multiplexer **185**A) in electrical communication with a voltage from the conductive plate structure **142**. In some embodiments, the conductive plate structure **142** may be configured to provide a negative voltage (e.g., a drain voltage V.sub.dd or a voltage source supply V.sub.ss) to the source region **188**A of multiplexers **185**A of unselected vertical stacks of memory cells **120**. In some embodiments, providing the voltage from the conductive plate structure **142** on the source regions **188**A may reduce interference from unselected memory cells **120** during operation of the vertical stacks of memory cells **120**, such as by providing the voltage of the conductive plate structure **142** to the conductive pillar structure **160** of unselected vertical stacks of memory cells **120**. In other words, the bleeder transistors **185**B are configured to electrically connect unselected conductive pillar structures **160** to their respective conductive plate structures **142** (e.g., ground structures, cell plates), which may be coupled to a negative voltage. In some embodiments, each vertical stack of memory cells **120** includes at least one (e.g., one) of the multiplexers **185**A and at least one (e.g., one) of the bleeder transistors **185**B.
(112) Each of the gate structures **189**, the first conductive interconnect structure **192**, the second conductive interconnect structures **195**, the third conductive interconnect structures **196**, the first routing structures **194**, the second routing structures **172**, and the third routing structures **173** may individually be formed of and include conductive material, such as one or more of the materials described above with reference to the global digit lines **108**. In some embodiments, the gate structures **189**, the first conductive interconnect structure **192**, the second conductive interconnect structures **195**, the third conductive interconnect structures **196**, the first routing structures **194**, the second routing structures **172**, and the third routing structures **173** are individually formed of and include tungsten. In other embodiments, the gate structures **189**, the first conductive interconnect structure **192**, the second conductive interconnect structures **195**, the third conductive interconnect structures **196**, the first routing structures **194**, the second routing structures **172**, and the third routing structures **173** are individually formed of and include copper.
(113) In some embodiments, each of the global digit lines **108**, the global digit line contact structures **162**, the gate structures **189**, the first conductive interconnect structure **192**, the second conductive interconnect structures **195**, the third conductive interconnect structures **196**, the first routing structures **194**, the second routing structures **172**, and the third routing structures **173** are formed in the seventh insulative material **187**.
(114) The seventh insulative material **187** may be formed of and include insulative material, such as one or more of the materials described above with reference to the first insulative material **114**. In some embodiments, the seventh insulative material **187** is formed of and includes silicon dioxide.
(115) Although the vertical stacks of memory cells **120** have been illustrated as comprising four (4)

levels of the memory cells **120**, the disclosure is not so limited. The vertical stacks of memory cells **120** may include a desired quantity of the levels of the memory cells **120**, such as within a range from thirty-two (32) of the levels of the memory cells **120** to two hundred fifty-six (256) of the levels of the memory cells **120**, as described above with reference to the quantity of the steps **175** of the staircase structures **174**.

(116) Although FIG. **1**J illustrates thirty-two (32) vertical stacks of memory cells **120** (e.g., eight (8) rows and four (4) columns of vertical stacks of memory cells **120**), the disclosure is not so limited, and the array region **101** may include greater than thirty-two vertical stacks of memory cells **120**. For example, the first microelectronic device structure **100** may include a greater quantity of columns of the vertical stacks of memory cells **120**, a greater quantity of the rows of the vertical stacks of memory cells **120**, or both.

(117) FIG. **2**A and FIG. **2**B are simplified partial cross-sectional views illustrating a microelectronic device **200** formed from the first microelectronic device structure **100** and a second microelectronic device structure **250** after attaching the second microelectronic device structure **250** to the first microelectronic device structure **100**. FIG. **2**A is a simplified cross-sectional view of the microelectronic device **200** and illustrates the same cross-sectional view of the first microelectronic device structure **100** illustrated in FIG. **1**K. FIG. **2**B is a simplified cross-sectional view of the microelectronic device **200** and illustrates the same cross-sectional view of the first microelectronic device structure **100** illustrated in FIG. **1**C.

(118) By way of non-limiting example, the second microelectronic device structure **250** may be attached to the first microelectronic device structure **100** by oxide-to-oxide bonding. In some such embodiments, an oxide material of the second microelectronic device structure **250** is brought into contact with an oxide material of the first microelectronic device structure **100** and the first microelectronic device structure **100** and the second microelectronic device structure **250** are exposed to annealing conditions to form bonds (e.g., oxide-to-oxide bonds) between the oxide material of the first microelectronic device structure **100** and the oxide material of the second microelectronic device structure **250**.

(119) The second microelectronic device structure **250** may include control logic devices (e.g., CMOS devices) and circuitry configured for effectuating control operations for the memory cells **120**. By way of non-limiting example, the second microelectronic device structure **250** may include one or more sub word line driver regions, one or more socket regions, and one or more additional CMOS regions including one or more of (e.g., all of) one or more sense amplifier devices (e.g., equalization (EQ) amplifiers, isolation (ISO) amplifiers, NMOS sense amplifiers (NSAs), PMOS sense amplifiers (PSAs)), column decoders, multiplexer control logic devices, sense amplifier drivers, main word line driver devices, row decoder devices, and row select devices.

(120) With reference to FIG. **2**A, the second microelectronic device structure **250** may include one or more sense amplifier device regions **202** vertically overlying (e.g., in the Z-direction) and within horizontal boundaries of the vertical stacks of memory cells **120**. The sense amplifier device regions **202** may include transistor structures in electrical communication with the global digit lines **108** by means of fourth conductive interconnect structures **204**. In some embodiments, the sense amplifier device regions **202** include sense amplifier devices (e.g., equalization (EQ) amplifiers, isolation (ISO) amplifiers, NMOS sense amplifiers (NSAs), PMOS sense amplifiers (PSAs)).

(121) With reference to FIG. **2**B, the second microelectronic device structure **250** may include one or more sub word line driver regions **206** vertically overlying (e.g., in the Z-direction) and within horizontal boundaries of the staircase structures **174**, such as vertically overlying the conductive contact structures **176** and the pad structures **178**. The sub word line driver regions **206** may include transistor structures in electrical communication with the conductive structures **132** by means of fifth conductive interconnect structures **208** that are in electrical communication with the pad structures **178**.

(122) Each of the fourth conductive interconnect structures **204** and the fifth conductive

interconnect structures **208** are individually formed of and include conductive material, such as one or more of the conductive materials described above with reference to the first conductive interconnect structures **192**. In some embodiments, the fourth conductive interconnect structures **204** and the fifth conductive interconnect structures **208** individually comprise tungsten. In other embodiments, the fourth conductive interconnect structures **204** and the fifth conductive interconnect structures **208** individually comprise copper.

(123) Thus, in accordance with some embodiments, a microelectronic device comprises vertical stacks of memory cells, each vertical stack of memory cells comprising a vertical stack of access devices, a vertical stack of capacitors horizontally neighboring the vertical stack of access devices, and a conductive pillar structure vertically extending through the vertical stack of access devices. The microelectronic device further comprises multiplexers and additional transistors vertically overlying the vertical stacks of memory cells, and global digit lines vertically overlying the multiplexer and the additional transistor.

(124) Furthermore, in accordance with additional embodiments of the disclosure, a microelectronic device comprises vertical stacks of dynamic random access memory (DRAM) cells, each DRAM cell comprising a storage device horizontally neighboring an access device, at least one multiplexer vertically overlying at least one of vertical stacks of DRAM cells, and at least one additional transistor structure vertically overlying the at least one of the vertical stacks of DRAM cells and horizontally neighboring the at least one multiplexer, the at least one multiplexer and the at least one additional transistor structure sharing a source region.

(125) Moreover, in accordance with some embodiments of the disclosure, a method of forming a microelectronic device comprises forming a conductive material in an opening vertically extending through vertical stacks of access devices horizontally neighboring one another to form a conductive pillar structure in electrical communication with the access devices of each of the vertical stack of access devices, forming an insulative liner material over the conductive material in the opening, removing a lower portion of the insulative liner material and the conductive material at the lower portion of the opening to form isolated conductive pillar structures individually in electrical communication with one of the vertical stacks of access devices, forming trenches through a stack structure of vertically alternating first materials and second materials on a side of each of the vertical stacks of access devices opposite the conductive pillar structure, forming vertical stacks of capacitor structures in the trenches to form vertical stacks of memory cells, each vertical stack of memory cells comprising a vertical stack of capacitor structures in electrical communication with a vertical stack of access devices, and forming a multiplexer and at least one additional transistor vertically over each of the vertical stack of memory cells.

(126) Structures, assemblies, and devices in accordance with embodiments of the disclosure may be included in electronic systems of the disclosure. For example, FIG. **3** is a block diagram of an illustrative electronic system **300** according to embodiments of disclosure. The electronic system **300** may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), portable media (e.g., music) player, a Wi-Fi or cellular-enabled tablet such as, for example, an iPad® or SURFACE® tablet, an electronic book, a navigation device, etc. The electronic system **300** includes at least one memory device **302**. The memory device **302** may comprise, for example, an embodiment of one or more of a microelectronic device structure, a microelectronic device structure assembly, a relatively larger microelectronic device structure assembly, and a microelectronic device previously described herein with reference to FIG. **1**A through FIG. **2**B. The electronic system **300** may further include at least one electronic signal processor device **304** (often referred to as a "microprocessor"). The electronic signal processor device **304** may, optionally, include an embodiment of one or more of a microelectronic device structure, a microelectronic device structure assembly, a relatively larger microelectronic device structure assembly, and a microelectronic device previously described herein with reference to FIG. **1**A through FIG. **2**B.

While the memory device **302** and the electronic signal processor device **304** are depicted as two (2) separate devices in FIG. **3**, in additional embodiments, a single (e.g., only one) memory/processor device having the functionalities of the memory device **302** and the electronic signal processor device **304** is included in the electronic system **300**. In such embodiments, the memory/processor device may include one or more of a microelectronic device structure, a microelectronic device structure assembly, a relatively larger microelectronic device structure assembly, and a microelectronic device previously described herein with reference to FIG. **1**A through FIG. **2**B. The electronic system **300** may further include one or more input devices **306** for inputting information into the electronic system **300** by a user, such as, for example, a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system **300** may further include one or more output devices **308** for outputting information (e.g., visual or audio output) to a user such as, for example, one or more of a monitor, a display, a printer, an audio output jack, and a speaker. In some embodiments, the input device **306** and the output device **308** may comprise a single touchscreen device that can be used both to input information to the electronic system **300** and to output visual information to a user. The input device **306** and the output device **308** may communicate electrically with one or more of the memory device **302** and the electronic signal processor device **304**.
(127) Thus, in accordance with embodiments of the disclosure, an electronic system comprises an input device, an output device, a processor device operably coupled to the input device and the output device, and a memory device operably coupled to the processor device. The memory device comprises a vertical stack of capacitor structures horizontally neighboring a vertical stack of access devices, a conductive plate structure in electrical communication with an electrode of each of the capacitor structures, a conductive pillar structure in electrical communication with the vertical stack of access devices, an additional vertical stack of capacitor structures comprising additional electrodes in electrical communication with the conductive pillar structure, the conductive plate structure horizontally intervening between the additional vertical stack of capacitor structures and the vertical stack of capacitor structures, and a multiplexer and a bleeder transistor vertically overlying the vertical stack of capacitor structures and the vertical stack of access devices.
(128) While certain illustrative embodiments have been described in connection with the figures, those of ordinary skill in the art will recognize and appreciate that embodiments encompassed by the disclosure are not limited to those embodiments explicitly shown and described herein. Rather, many additions, deletions, and modifications to the embodiments described herein may be made without departing from the scope of embodiments encompassed by the disclosure, such as those hereinafter claimed, including legal equivalents. In addition, features from one disclosed embodiment may be combined with features of another disclosed embodiment while still being encompassed within the scope of the disclosure.

## Claims

1. A microelectronic device, comprising: vertical stacks of memory cells, each vertical stack of memory cells comprising: a vertical stack of access devices; a vertical stack of capacitors horizontally neighboring the vertical stack of access devices; and a conductive pillar structure vertically extending through the vertical stack of access devices; multiplexers and additional transistors vertically overlying the vertical stacks of memory cells; and global digit lines vertically overlying the multiplexer and the additional transistors.
2. The microelectronic device of claim 1, wherein each multiplexer individually shares a source region with one of the additional transistors.
3. The microelectronic device of claim 1, further comprising a conductive plate structure vertically extending through the vertical stacks of memory cells and neighboring the vertical stacks of capacitors of the vertical stacks of memory cells.

4. The microelectronic device of claim 3, wherein pairs of the additional transistors are in electrical communication with the conductive plate structure.
5. The microelectronic device of claim 1, wherein: one of the multiplexers vertically overlies each vertical stack of memory cells; and two multiplexers are located between horizontally neighboring conductive pillar structures of horizontally neighboring vertical stacks of memory cells.
6. The microelectronic device of claim 1, wherein horizontally neighboring multiplexers are individually in electrical communication with a respective conductive pillar structure vertically extending through a respective vertical stacks of memory cells.
7. The microelectronic device of claim 1, further comprising a vertical stack of conductive structures intersecting multiple of the vertical stacks of memory cells.
8. The microelectronic device of claim 7, wherein each conductive structure of the vertical stack of conductive structures is configured to be in electrical communication with an access device of the vertical stacks of memory cells intersected by the vertical stack of conductive structures.
9. The microelectronic device of claim 1, further comprising global digit line contact structures individually in electrical communication with a drain region of one of the multiplexers and one of the global digit lines.
10. The microelectronic device of claim 1, wherein drain regions of pairs of the additional transistors horizontally neighboring one another are in electrical communication with each other.
11. The microelectronic device of claim 1, wherein a source region of each of the multiplexers is in electrical communication with one of the conductive pillar structures.
12. A microelectronic device, comprising: vertical stacks of dynamic random access memory (DRAM) cells, each DRAM cell comprising a storage device horizontally neighboring an access device; at least one multiplexer vertically overlying at least one of the vertical stacks of DRAM cells; and at least one additional transistor structure vertically overlying the at least one of the vertical stacks of DRAM cells and horizontally neighboring the at least one multiplexer, the at least one multiplexer and the at least one additional transistor structure sharing a source region.
13. The microelectronic device of claim 12, further comprising global digit lines vertically overlying the at least one multiplexer.
14. The microelectronic device of claim 12, wherein the source region is in electrical communication with a conductive pillar structure vertically extending through the at least one of the vertical stacks of DRAM cells.
15. The microelectronic device of claim 12, wherein a drain region of the at least one multiplexer is in electrical communication with a global digit line.
16. The microelectronic device of claim 12, wherein a drain region of the at least one additional transistor structure is in electrical communication with a conductive plate structure.
17. The microelectronic device of claim 12, wherein a drain region of the at least one multiplexer is horizontally aligned with a drain region of the at least one additional transistor.
18. The microelectronic device of claim 17, wherein the drain region of the at least one multiplexer and the drain region of the at least one additional transistor are horizontally aligned with a global digit line in electrical communication with the at least one multiplexer.
19. The microelectronic device of claim 17, wherein the source region is horizontally offset from the drain region of the at least one multiplexer and the drain region of the at least one additional transistor.
20. The microelectronic device of claim 12, wherein a gate structure of the at least one multiplexer extends in a horizontal direction substantially perpendicular to word lines configured to be in electrical communication with the access devices of the DRAM cells.
21. A method of forming a microelectronic device, the method comprising: forming a conductive material in an opening vertically extending through vertical stacks of access devices horizontally neighboring one another to form a conductive pillar structure in electrical communication with the access devices of each of the vertical stack of access devices; forming an insulative liner material

over the conductive material in the opening; removing a lower portion of the insulative liner material and the conductive material at the lower portion of the opening to form isolated conductive pillar structures individually in electrical communication with one of the vertical stacks of access devices; forming trenches through a stack structure of vertically alternating first materials and second materials on a side of each of the vertical stacks of access devices opposite the conductive pillar structure; forming vertical stacks of capacitor structures in the trenches to form vertical stacks of memory cells, each vertical stack of memory cells comprising a vertical stack of capacitor structures in electrical communication with a vertical stack of access devices; and forming a multiplexer and at least one additional transistor vertically over each of the vertical stack of memory cells.

22. The method of claim 21, wherein forming vertical stacks of capacitor structures in the trenches to form vertical stacks of memory cells comprises: forming a first electrode material in the trenches; forming a dielectric material over the first electrode material; and forming a second electrode material over the dielectric material.

23. The method of claim 22, further comprising forming a conductive plate structure over the second electrode material in the trenches.

24. The method of claim 23, wherein forming at least one additional transistor comprises forming a drain region of the at least one additional transistor in electrical communication with the conductive plate structure.

25. The method of claim 21, wherein forming a multiplexer comprises forming a drain region of the multiplexer in electrical communication with a global digit line vertically overlying the multiplexer.

26. The method of claim 21, wherein forming a multiplexer and at least one additional transistor comprises forming a source region of the multiplexer shared with the at least one additional transistor.

27. An electronic system, comprising: an input device; an output device; a processor device operably coupled to the input device and the output device; and a memory device operably coupled to the processor device and comprising: a vertical stack of capacitor structures horizontally neighboring a vertical stack of access devices; a conductive plate structure in electrical communication with an electrode of each of the capacitor structures; a conductive pillar structure in electrical communication with the vertical stack of access devices; an additional vertical stack of capacitor structures comprising additional electrodes in electrical communication with the conductive pillar structure, the conductive plate structure horizontally intervening between the additional vertical stack of capacitor structures and the vertical stack of capacitor structures; and a multiplexer and a bleeder transistor vertically overlying the vertical stack of capacitor structures and the vertical stack of access devices.

28. The electronic system of claim 27, further comprising global digit lines vertically overlying the multiplexer and the bleeder transistor.

29. The electronic system of claim 28, wherein: a drain region of the multiplexer is in electrical communication with one of the global digit lines; and a source region of the multiplexer is in electrical communication with the conductive pillar structure.

30. The electronic system of claim 27, wherein the multiplexer comprises a source region shared with the bleeder transistor.